META GRAPH API MASTER GUIDE

Centralized Setup, Permanent System Tokens & 50+ Account Architecture

OmniStudio AI Engine Documentation • Targeted for Notion & Production Engineering

01 Architecture & Centralized Scaling Model

OmniStudio AI publishes automated video reels, AI-branded framing templates, and dynamic Instagram Stories directly through the official Meta Graph API (v21.0). When scaling up to 50+ accounts, creating separate Meta developer apps for each account creates unmanageable overhead. Instead, we implement a Centralized Hub-and-Spoke Architecture.

Component	Role in Ecosystem	Centralized Scope
1 Developer App	Registered in Meta Developers; manages Graph API calls	Shared across all 50+ accounts
1 Business Portfolio	Meta Business Manager holding all company brand pages	Holds 50 Facebook Pages & 50 IG Accounts
1 System User	Administrative service user (omni-bot-admin)	Granted Full Control on all 50 assets
1 Permanent Token	Never-expiring access token generated via System User	One master credential runs the entire fleet

& þ Permanent Token vs Standard User Token

Standard User Tokens generated via Graph API Explorer expire after 60 days. System User Tokens created inside Meta Business Settings have NO expiration date ("Never"). You set it once, and the system runs continuously without manual token renewal.

02 Instagram & Facebook Foundation Setup

Meta Graph API strictly requires that every automated Instagram account is converted to a Professional Account and linked to an associated Facebook Page.

Step 2.1: Convert Instagram to a Professional Account

- %¸ Open Instagram app > Profile > Hamburger Menu ("a) > Settings & Privacy.
- %¸ Scroll to "For professionals" > Tap "Account type and tools" > Switch to Professional Account.
- %¸ Select category (e.g. Video Creator, Gaming, or Entertainment). Choose Creator or Business.

Step 2.2: Create a Dedicated Facebook Page

- %¸ Go to https://www.facebook.com/pages/create on desktop.
- %¸ Set Page Name matching your brand niche (e.g. "GTA 6 Updates 007" or "Movie Recommendations").
- %¸ Select category (Media/News Company or Video Creator) and click Create Page.

Step 2.3: Link Instagram to the Facebook Page

- %¸ Switch profile into the newly created Facebook Page.
- %¸ Go to Settings & Privacy > Settings > Linked Accounts > Select Instagram.
- %¸ Click Connect Account > Enter Instagram credentials > Confirm.
- %¸ Toggle ON "Allow access to Instagram messages in Inbox". Status should show Connected.

03 Meta for Developers Account & App Creation

Step 3.1: Register as a Meta Developer

- Navigate to https://developers.facebook.com/ and log into your primary Facebook account.
- Click Get Started > Accept Meta Developer Terms > Complete phone/email verification.

Step 3.2: Create the Master Application

- Go to https://developers.facebook.com/apps/ and click Create App.
- Step 1 ("What do you want your app to do?"): Select "Other" > Click Next.
- Step 2 ("Select an app type"): Select "Business" > Click Next.
- App Display Name: Set to "OmniStudio Automation Engine". Add your active Contact Email.
- Business Account: Connect to your Meta Business Portfolio > Click Create App.

Step 3.3: Configure Products

- In the App Dashboard, scroll to "Instagram Graph API" and click Set Up.
- Under App Roles, verify your developer account is listed as Administrator.

04 System User Setup & Generating Permanent Token

This is the critical step to achieve zero-maintenance automation. We create an Admin System User inside Meta Business Manager and generate an access token with permanent validity.

Step 4.1: Open Business Settings

- Navigate to Meta Business Settings at https://business.facebook.com/settings/.
- Ensure all brand Facebook Pages and Instagram accounts are added under Accounts.

Step 4.2: Create the Admin System User

- In the left sidebar, expand "Users" and click "System Users".
- Click "Add" > Set System Username: "omni-bot-admin".
- Crucial: Set System User Role to "Admin" (Employee role cannot publish content). Click Create.

Step 4.3: Assign All Brand Assets to System User

- Select "omni-bot-admin" > Click "Assign Assets".
- Under Pages: Select all Facebook Pages > Toggle "Full Control (Manage Page)" to ON.
- Under Instagram Accounts: Select all IG Accounts > Toggle "Full Control" to ON.
- Under Apps: Select "OmniStudio Automation Engine" > Toggle "Full Control" to ON > Click Save.

Step 4.4: Generate Permanent Token with Required Scopes

- Click "Generate New Token" on the omni-bot-admin panel.
- Select App: "OmniStudio Automation Engine".
- Token Expiration: Select "Never" (Permanent).
- Check the following required permissions (Scopes):

- **instagram_basic:** Allows reading basic IG account profile and media IDs
- **instagram_content_publish:** MANDATORY — Allows uploading video reels and publishing stories
- **instagram_manage_comments:** Allows AI comment replying and engagement tracking
- **instagram_manage_insights:** Allows fetching virality metrics, view counts, and retention
- **pages_show_list:** Allows querying the list of managed Facebook Pages

- **pages_read_engagement:** Enables reading interaction data and page associations
- **pages_manage_posts:** Allows feed video container creation and webhook management
- **business_management:** Enables multi-account portfolio governance under one token

> **& þ Save the Token Immediately**
> Meta displays this permanent token ONCE upon creation. Copy it and store it in your password manager immediately. It will begin with "EAAG..." and is approximately 180-220 characters long.

05 Extracting Account IDs & Token Debugging

To target an account during an automated run, the agent requires the 17-digit Instagram Business Account ID (ig_user_id).

Step 5.1: Extract ig_user_id via Meta Graph API Explorer

%¸ Open https://developers.facebook.com/tools/explorer/.
%¸ Paste your Permanent Token into the "Access Token" input field.
%¸ Enter the following query in the request bar and click Submit:

```
GET /me/accounts?fields=id,name,instagram_business_account{id,username}
```

The response will list all managed accounts. Locate the 17-digit ID:

```
{
  "data": [
    {
      "id": "104829104810294", // Facebook Page ID
      "name": "GTA 6 Updates 007",
      "instagram_business_account": {
        "id": "17841428668115319", // <--- THIS IS YOUR ig_user_id
        "username": "gta6_updates_007"
      }
    }
  ]
}
```

Step 5.2: Verify Token Health in Access Token Debugger

%¸ Visit https://developers.facebook.com/tools/debug/accesstoken/.
%¸ Paste your token and click Debug. Confirm: Type = SYSTEM_USER, Expires = Never.

06 Connecting to OmniStudio AI & GitHub Actions

To deploy the agent locally or to 24/7 cloud runners, populate the following credentials:

1. Environment Variables (.env)

```
# Meta Graph API Master Credentials
INSTAGRAM_ACCESS_TOKEN="EAAG..."
INSTAGRAM_PAGE_ID="17841428668115319"
FB_PAGE_ID="104829104810294"

# Telegram Real-time Publishing Notifications
TELEGRAM_BOT_TOKEN="123456789:ABCdefGhIJKlmNoPQRstuVWxYz"
TELEGRAM_ADMIN_CHAT_ID="5443723531"

# Instagram Scout Burner Cookie (Bypasses Datacenter Login Wall)
INSTAGRAM_SESSION_ID="YOUR_BURNER_SESSION_ID"
```

2. GitHub Actions Secrets (For 24/7 Cloud Automation)

Go to GitHub Repo > Settings > Secrets and variables > Actions > Add Secrets: INSTAGRAM_ACCESS_TOKEN, INSTAGRAM_PAGE_ID, TELEGRAM_BOT_TOKEN, TELEGRAM_CHAT_ID, and INSTAGRAM_SESSION_ID.

07 Scaling to 50+ Accounts: 3-Minute Onboarding SOP

When adding account #3, #10, or #50, NEVER create a new Meta Developer App. Use this 3-minute onboarding procedure:

Step 1: Create Account & Link

- Convert new IG to Professional > Create FB Page > Link IG to FB Page.

Step 2: Add to Business Manager

- Go to business.facebook.com/settings/pages > Add Page.

Step 3: Assign to System User

- Go to Users > System Users > omni-bot-admin > Assign Assets > Grant Full Control on new Page.

Step 4: Query New ig_user_id

- Run GET /me/accounts in Graph Explorer to get the 17-digit ID.

Step 5: Register in SQLite Database

- Add a single row to the pages table in database.sqlite (or via the web dashboard):

```
INSERT INTO pages (name, handle, ig_user_id, post_mode, is_active, daily_quota, access_token)
VALUES ('Crypto Alpha', '@crypto_alpha_007', '17841499999999999', 'auto', 1, 7, 'EAAG...');
```

Step 6: Define Competitors

- Add competitor creator handles into config/competitors.json.

08 Troubleshooting & Error Matrix

Error Code	Root Cause	Immediate Solution
Error 190	Token expired or revoked	Check Token Debugger. Generate System User token with "Never" expiry
Error 200 / 10	Missing publish permission	Assign asset to omni-bot-admin with Full Control; verify scopes.

Error 36003

Invalid video aspect ratio

OmniStudio FFmpeg helper automatically transcodes to 1080x1920.

302 Redirect

IG Datacenter login wall

Update INSTAGRAM_SESSION_ID secret with fresh burner scout cookie.